

ReactJS HOL - 3

Objective

Create a React application named scorecalculatorapp using a functional component named CalculateScore to calculate and display the average score of a student with CSS styling.

Theory

Explain React Components

React components are reusable building blocks used to create the user interface. They make applications modular, maintainable, and easier to develop.

Difference Between React Components and JavaScript Functions

React components return JSX and build the UI, whereas JavaScript functions perform general programming tasks and return values.

Types of Components

React has two main component types: Functional Components and Class Components.

Explain Class Component

A class component extends `React.Component` and contains a `render()` method. It can manage state and lifecycle methods.

Explain Function Component

A functional component is a JavaScript function that returns JSX. It is simple, reusable, and supports Hooks.

Define Component Constructor

The constructor initializes the state and binds methods in class components before rendering.

Define render() Function

The `render()` method returns the JSX that React displays on the browser and is called whenever state or props change.

Prerequisites

- Node.js
- npm
- Visual Studio Code

Implementation Steps

1. Create the project: `npx create-react-app scorecalculatorapp`
2. Create Components and Stylesheets folders under src.
3. Create CalculateScore.js functional component.
4. Create mystyle.css and add styles.
5. Import CalculateScore into App.js and pass Name, School, total and goal as props.
6. Run the application using `npm start`.
7. Verify the output in the browser.

CalculateScore.js

```
import '../Stylesheets/mystyle.css';

const percentToDecimal=(decimal)=> (decimal.toFixed(2)+'%');
const calcScore=(total,goal)=> percentToDecimal(total/goal);

export const CalculateScore=({Name,School,total,goal})=>{
  <div className='formatstyle'>
    <h1><font color='Brown'>Student Details:</font></h1>
    <div className='Name'><b>Name:</b> {Name}</div>
    <div className='School'><b>School:</b> {School}</div>
    <div className='Total'><b>Total:</b> {total} Marks</div>
    <div className='Score'><b>Score:</b> {calcScore(total,goal)}</div>
  </div>
);
```

mystyle.css

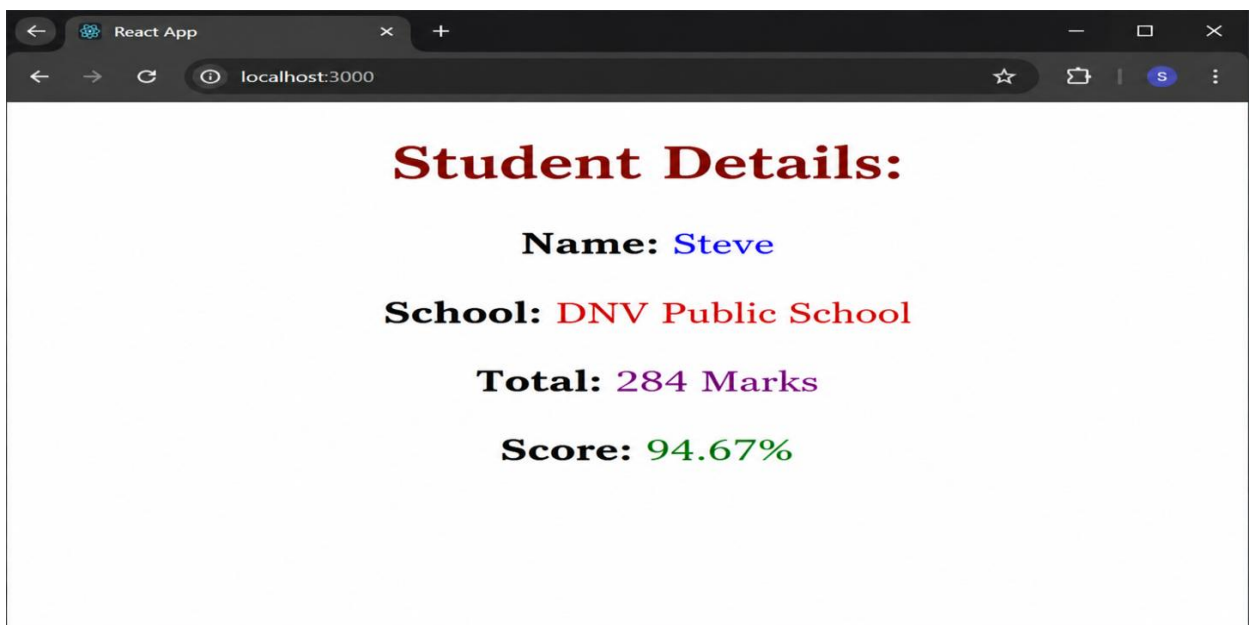
```
.Name{font-weight:300;color:blue;}
.School{color:crimson;}
.Total{color:darkmagenta;}
.Score{color:forestgreen;}
.formatstyle{text-align:center;font-size:large;}
```

App.js

```
import {CalculateScore} from './src/components/CalculateScore';

function App(){
  return(
    <div>
      <CalculateScore
        Name={'Steve'}
        School={'DNV Public School'}
        total={284}
        goal={3}
      />
    </div>
  );
}
export default App;
```

Output Screenshot:



Learning Outcome

Successfully created a React functional component, applied CSS styling, passed data using props, and rendered the calculated student score.